

■ BusinessNext

AWS Cost Estimator

Pricing & Infrastructure Report

HDFC

Mode: On-Premise — Client Hosted

Date: 09 April 2026

Inflation: 4% / yr

AWS Region: us-east-1

Executive Summary

High-level cost overview and infrastructure snapshot

4	48	88	96
Worker Nodes	vCPUs	RAM (GB)	DB RAM (GB)
Total compute nodes	Total virtual CPUs	Total memory	Database memory

Project Details

Customer / Bank	HDFC
Client Mode	On-Premise — Client Hosted
Database Type	SQL Server
Total Named Users	5,000
Concurrent Users	1,500 (30% of named)
Mobile Users	750
Worker Nodes	4
Total vCPUs (worker)	48
Total RAM GB (worker)	88 GB
Data Size	100 GB
Estimate Date	09 April 2026

Node Distribution

AI-assisted infrastructure architecture

Total Worker Nodes	4
Total DB Nodes	13
Fixed / Managed Roles	7
Confidence	High
Distribution Method	Groq LLM + Rule-based

Category	Role / Service	Nodes	Instance Fam.	vCPU/node	RAM GB/node	Model
Kubernetes Cluster	Worker Nodes Linux/RHEL (Web, Mobile, WebAPI)	2	Compute Intensive	16	32	Reserved 3 Yr
Kubernetes Cluster	Worker Nodes Linux/RHEL for Graphana & Prometheus	1	Compute Intensive	16	32	Reserved 3 Yr
Kubernetes Cluster	Worker Nodes Linux/RHEL for EFK	1	Compute Intensive	16	32	Reserved 3 Yr
SQL Server Database	Windows Nodes – Database server (Primary)	2	Memory Intensive	32	128	None
SQL Server Database	Nodes – Cluster (Always On / AD / Witness)	4	Memory Intensive	2	8	None
SQL Server Database	Storage: SAN (Primary SQL Server DB)	2	10K IOPS	—	—	None
S3	S3 Storage + Replication	1	Cloud Storage Service	—	—	On Demand
SQL Server Database – Reporting	Windows Nodes – Database server (Reporting)	1	Memory Intensive	32	128	None
SQL Server Database – Reporting	Storage: SAN (Reporting SQL Server DB)	1	10K IOPS	—	—	None
Security & Compliance	WAF + AWS Shield Advanced	1	Managed Service	—	—	On Demand
Security & Compliance	CloudTrail + Config Audit Logs	1	Managed Service	—	—	On Demand
Kubernetes Cluster	Cloud Managed K8s	1	K8s Service	—	—	None
Caching & Session	Elastichache Service	2	Memory Optimized	4	16	Reserved 3 Yr
Infrastructure & Monitoring	Load Balancer (Internal + External)	2	ALB	—	—	On Demand
Infrastructure & Monitoring	Bastion Host	1	General Purpose	2	4	Reserved 3 Yr
Infrastructure & Monitoring	NAT Gateway	2	NAT	—	—	On Demand
Infrastructure & Monitoring	Image Registry (ECR)	1	Managed Service	—	—	On Demand
Infrastructure & Monitoring	Back Up - DB & Infra Logs	1	Cloud Storage Service	—	—	On Demand

Environment Pricing

Pre-Prod / SIT / UAT and DR environment costs

Environments: Pre-Prod, SIT, UAT

Environments Selected	3 — Pre-Prod, SIT, UAT
Base Cost per Env / mo	\$1,133
Base Cost per Env / yr	\$13,591
Calculation	3 envs × \$1,133/mo = \$3,398/mo total

Monthly	Annual	5-Year
\$3,398	\$40,774	\$0

Category	Role	Nodes	Instance	\$/month
Kubernetes	Cloud Managed K8s	1	EKS	\$73
Kubernetes	Worker Nodes (Web, Mobile, WebAPI)	2	m5.4xlarge	\$1,169
Kubernetes	Worker Nodes (Graphana & Prometheus) – optional	1	m5.2xlarge	\$321
Kubernetes	Worker Nodes (EFK) – optional	1	m5.4xlarge	\$602
S3	S3 Storage + Replication	1	S3	\$34
Caching	Caching Service (ElastiCache)	1	cache.r6g.large	\$121
Infrastructure	Load Balancer (Internal + External)	1	ALB	\$16
Infrastructure	NAT Gateway	2	NAT	\$32
Infrastructure	Bastion Host	1	m5.large	\$94
Infrastructure	Image Registry (ECR)	1	ECR	\$50
Backup	Back Up – DB & Infra Logs	1	S3	\$34
PGSQL DB	—	—	db.r5.2xlarge	\$850

Disaster Recovery (DR) Environment

Monthly	Annual	5-Year
\$7,719	\$92,625	\$521,756

Category	Role	Nodes	Instance	\$/month
Kubernetes	Cloud Managed K8s (DR)	1	EKS	\$73
Kubernetes	Worker Nodes (Web, Mobile, WebAPI) – DR	4	m5.4xlarge	\$2,324
Kubernetes	Worker Nodes (Graphana) – DR	2	m5.2xlarge	\$643
Kubernetes	Worker Nodes (EFK) – DR	2	m5.2xlarge	\$724
S3	S3 Storage + Replication – DR	1	S3	\$184

Category	Role	Nodes	Instance	\$/month
Caching	Elastichache Service – DR	2	cache.r6g.large	\$242
Infrastructure	Load Balancer – DR	2	ALB	\$33
Infrastructure	NAT Gateway – DR	2	NAT	\$32
Infrastructure	Bastion Host – DR	1	m5.large	\$94
Infrastructure	Image Registry (ECR) – DR	1	ECR	\$51
Backup	Back Up – DR	1	S3	\$118
Managed DB	—	—	db.r5.4xlarge	\$3,200

Notes & Assumptions

Detailed pricing basis, scope, and validity

■ Pricing Validity — 15 Days Only

This estimate is valid for **15 calendar days** from the date of issue (**09 April 2026**). AWS On-Demand prices, instance availability, and currency exchange rates are subject to change without notice. A revised estimate must be requested if requirements change or the validity period lapses.

Pricing & Infrastructure Assumptions

AWS Pricing Basis	All prices are AWS On-Demand, Linux/UNIX, shared tenancy. No Reserved Instances or Savings Plans applied. Switching to 1-Year Reserved pricing typically reduces compute costs by 30-40%.
Region	Estimate uses region: us-east-1. Costs vary by region — a regional multiplier is applied to the base us-east-1 rate.
Inflation Rate	4% per annum applied to all compute and managed-service costs from Year 2 onwards, reflecting typical AWS price adjustments and currency exposure.
High Availability (AZ)	Production and DR environments are architected as Multi-AZ (High Availability). Pre-Prod/SIT/UAT are Single-AZ to optimize costs while maintaining functional parity.
Database Licensing	Postgres is Open-Source (No license cost). SQL Server and Oracle are priced on an AWS RDS 'License Included' basis. Enterprise/BYOL discounts are not applied.
Fallback Rates	If the live AWS Pricing API is unavailable, hardcoded March 2026 on-demand estimates are used. API-sourced prices are marked with a checkmark; estimated prices are noted as estimated.
Node Distribution	Infrastructure sizing computed by a rule-based algorithm, optionally enhanced by Groq LLM (llama-3.3-70b-versatile) at temperature=0. SHA-256 caching prevents redundant API calls.
Named & Concurrent Users	Concurrent users are fixed at 30% of Total Named Users for every forecast year. Year-over-Year growth is applied per the rates selected in the estimator tool.
Security & Compliance Costs	Fixed INR-converted estimates included: Airtel SOC \$500/mo, SOC Machines Rs.30,000/yr, Req 4 Antivirus Rs.2,600/yr, Req 6 Data Discovery Rs.4,000/yr. These are not subject to AWS price changes.
Pre-Prod / SIT / UAT	Environments sized at ~40% of Production infrastructure. PostgreSQL uses self-hosted EC2 (Patroni HA). SQL Server / Oracle use AWS RDS Managed. Costs are additive and not included in the Production PUPM.
Disaster Recovery (DR)	DR sized using a Pilot Light / Warm Standby model at ~50% of Production compute. Active-Active DR would approximately double the DR cost shown.
GCP Pricing	GCP prices are hardcoded us-central1 On-Demand Compute Engine estimates (March 2026). GCP Committed Use Discounts (1-Year CUD) typically reduce costs by ~37%.
PUPM Formula	$PUPM = \frac{\text{Discounted Annual Cost}}{12 / \text{Named Users}}$ Discounted cost = Total AWS Cost x (1 - 11% discount). Total includes production, DR, Pre-Prod, buffer (5%), Business Support (3.91%), and one-time migration costs in Year 1.
Managed Services	Managed Services fees (30% of AWS annual total) cover platform monitoring, patching, incident management, and lifecycle support by the BusinessNext operations team.

Scope of This Estimate

Included	Production cloud infrastructure (compute, database, storage, networking)
Included	Pre-Production / SIT / UAT environments (if selected)
Included	Disaster Recovery environment (if selected)
Included	BusinessNext platform licensing and managed services
Included	One-time migration, performance testing, and managed-service setup (Year 1)
Excluded	Application development or customisation effort
Excluded	End-user device provisioning or on-site hardware
Excluded	Third-party SaaS subscriptions (e.g. Salesforce, Twilio, Okta)
Excluded	Data centre co-location or physical network charges
Excluded	Regulatory compliance certification fees

Legal Disclaimer

This document is prepared by BusinessNext for the exclusive use of the named client organisation. All cost figures are estimates based on current AWS/GCP published pricing and the inputs provided at the time of generation. Actual costs may vary depending on usage patterns, reserved-instance commitments, enterprise discount programmes, and any changes to cloud provider pricing.

This estimate does **not** constitute a binding commercial offer or contract. A formal Statement of Work (SOW) and Order Form must be executed prior to commencement of any services. BusinessNext reserves the right to revise this estimate upon receipt of updated requirements or upon expiry of the 15-day validity period.

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